AN EMPIRICAL STUDY IN QUANTITATIVE FINANCE

A study of momentum in two equity markets

Residual Momentum Across Markets

The hidden fragility of momentum in the United States

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ABSTRACT

Momentum means buying shares that have been rising and selling shares that have been falling. Residual momentum is a cleaned up version of it. Before ranking the shares, it removes the part of each share's movement that came from the whole market, and keeps only what the company did on its own. It is widely believed to be the better, steadier version. This study asks a simple question: does it really work better in every market? We test it in the United States and in Europe. It works in Europe. In the United States it does not. We then ask why, and we test three explanations one at a time. The first is that the United States simply had more years of data. The second is that the difference is in the market itself. The third is that we removed the market in too simple a way. Only the second explanation survives. Over the same years, removing the market leaves a healthy signal in Europe and a losing, unreliable one in the United States, and this stays true even when we remove the market more thoroughly. The difference is not the time period and not the method. It is the market.

1

The Research Question

Does the cleaned up version of momentum work in every market, or only in some?

Momentum is one of the oldest and most reliable ideas in investing (Jegadeesh and Titman, 1993). Shares that have gone up over the past year tend to keep going up for a while, and shares that have gone down tend to keep going down. So an investor who buys the recent winners and sells the recent losers tends, on average, to make money. This pattern has shown up across many countries and many decades (Rouwenhorst, 1998; Asness, Moskowitz, and Pedersen, 2013).

A few years ago researchers suggested an improvement. They pointed out that part of a share's recent rise was never really about the company. When the whole market goes up, it carries almost every share with it, the strong and the weak alike. So they built a cleaned up version, called residual momentum, that first removes the market's push and then ranks each company only on what it did on its own. The promise was a cleaner, calmer, more dependable signal.

This study asks one plain question. Does that cleaned up version actually work better everywhere, or only in the markets where it was first tested? We answer it by building the same strategies in two of the world's largest markets, the United States and Europe, and comparing them carefully. To see why the cleaned up version is expected to be better, and therefore why it is surprising when it fails, we start with a short background.

2

A Short Background

What momentum is, and why the cleaned up version is trusted.

Ordinary momentum simply buys recent winners and sells recent losers (Jegadeesh and Titman, 1993). Its profits are not fully explained by the usual sources of risk (Fama and French, 1993; Carhart, 1997), which is why it is treated as a force of its own. It also has a well known weakness. Every so often it crashes hard. This usually happens when a falling market suddenly turns upward and the beaten down shares that the strategy is betting against come roaring back (Daniel and Moskowitz, 2016).

Residual momentum was the proposed fix (Grundy and Martin, 2001; Blitz, Huij, and Martens, 2011). The idea is to remove the part of each company's return that came from the market and a few other broad forces, and to keep only what is left, the part that belongs to the company itself. A simple way to picture it is an exam grade adjusted for how hard the exam was. Seventy out of a hundred on a very hard paper is more impressive than seventy on an easy one. Residual momentum makes that kind of fairness adjustment for every company, every month. Those earlier studies, carried out mainly in the United States, found it worked well. What nobody had checked carefully is whether it works equally well in different markets, and that gap is what this study fills.

3

The Data

Two markets, and a deliberate choice to keep the companies that failed.

We use monthly data for the companies in two well known indices. The S&P 500 stands in for the United States and the Stoxx 600 stands in for Europe.

One choice matters more than any other. We did not only use the companies that are in each index today. We used every company that was ever in each index during the study, including the ones that later shrank, merged, or went bust. If you leave out the failures, the past looks far safer and more profitable than it really was, because the disasters quietly vanish from the record. Keeping them in keeps the study honest.

Table 1. The two markets we study

Market	Years of data	Companies
United States, S&P 500	1990 to 2018	1,355
Europe, Stoxx 600	2002 to 2018	1,255

Why this matters. *The United States data start earlier, simply because the European records begin later. That difference in length looks harmless here, but it becomes the centre of the first test later on.*

4

How We Built and Measured the Strategies

Three strategies, written out plainly, and the four numbers we judge them by.

We build three strategies in each market. Each one is rebuilt every month, and each one buys the strongest tenth of companies and sells the weakest tenth.

The first is **plain momentum**. Each month we score every company by how much it rose or fell over the past year, but we skip the most recent month, because very recent moves often reverse and would mislead us. In symbols, the score for company i in month t is:

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M(i,t) = (1 + r(i,t-12)) x (1 + r(i,t-11)) x ... x (1 + r(i,t-2))
-  1
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Reading that line in plain words: M(i,t) is the momentum score for company i in month t. The letter r is a monthly return, so r(i,t-12) is the company's return twelve months ago. The line multiplies the returns from twelve months ago up to two months ago, which gives the company's total gain over that stretch. We buy the highest scores and sell the lowest.

The second is **residual momentum**. It does the same ranking, but first it removes the market's push. This happens in two steps. In step one, for each company we run a simple fit over the last thirty six months that splits the company's return into the part explained by three broad forces and the part left over:

```
r(i,s) - rf(s) = a + b1 x Market + b2 x Size + b3 x Value + e(i,s)
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Reading that line: the left side is the company's return above the safe, risk free rate. Market, Size and Value are the three broad forces, explained just below. The numbers b1, b2 and b3 measure how strongly each force moved this particular company. And e, the final term, is the residual, the part of the return that none of the three forces can explain. That leftover is the company's own performance, which is exactly what we want to rank on.

The three forces we remove, known together as the Fama and French three factor model, are these. Market is the whole market rising or falling, which lifts or drops almost everything at once. Size is the tendency for small companies to behave differently from large ones. Value is the tendency for cheap companies, priced low relative to their books, to behave differently from expensive ones. Removing all three leaves only what the company itself did.

In step two, we take that leftover for the last twelve months, skipping the most recent month, and turn it into a score:

```
RM(i,t) = (average of the leftover e over those months) / (how much the leftover wobbled)
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Here RM(i,t) is the residual momentum score. The top of the fraction is the company's average leftover return. The bottom is how much that leftover bounced around from month to month. Dividing one by the other rewards steady, dependable strength and punishes lucky, jumpy gains.

The third is **volatility targeted momentum**. It keeps the plain momentum picks but changes how much money goes in, depending on how calm or wild markets have been:

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w(t) = (a steady target risk level) / (how wild the market has been recently)
```

In that line, w(t) is how large the position is in month t. The target risk level is a fixed amount of risk we choose to aim for. When markets are calm, the bottom of the fraction is small, so w is large and we invest more. When markets are stormy, the bottom is large, so w is small and we pull back. The aim is to be holding less right when a crash is most likely.

We judge all three on the same four numbers;

The annualised return is the average gain per year.

The annualised volatility is how much the monthly returns bounced around, written as a yearly figure; a smaller number means a steadier ride.

The Sharpe ratio is the return divided by the volatility, so it is the reward earned for each unit of risk, and higher is better.

The maximum drawdown is the deepest fall from a previous high, which tells you the worst stretch an investor would have had to live through.

A few practical details make the study reproducible. All returns are monthly **total returns**, which include dividends, not only price changes. The companies come from a single **survivorship free** dataset of index members that keeps every firm that was ever in each index, including those that later left, merged, or failed. Each month the strongest and weakest tenths are held in **equal weight**, so every chosen company counts the same, and the portfolios are rebuilt fresh each month rather than overlapped. When a company has no return in a given month, for example before it joined the index or after it left, it is left out of that month's ranking rather than filled in with a guess.

5

The First Look

Each market measured over its own full history.

We start by measuring each market over all the years it has. Plain momentum makes money in both. It returns 4.4 percent a year in the United States and 14.9 percent in Europe. So far the two markets agree, and Europe is simply the stronger of the two.

Table 2. How each strategy did over its own full history

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
United States, plain	4.4%	29.4%	0.31	81.9%
United States, residual	1.6%	18.3%	0.20	74.0%
United States, volatility targeted	8.7%	14.4%	0.65	35.3%
Europe, plain	14.9%	23.4%	0.74	74.1%
Europe, residual	5.2%	9.1%	0.60	26.2%
Europe, volatility targeted	16.6%	13.7%	1.20	27.3%

Reading the numbers. *Annualised return is the average yearly gain. Annualised volatility is how much the returns bounced around, as a yearly figure; lower is steadier. The Sharpe ratio is return divided by volatility, so it is reward for each unit of risk. Maximum drawdown is the deepest fall from a previous high. Bigger return and Sharpe are better, smaller volatility and drawdown are better.*

Now look at the cleaned up version, residual momentum, in each market. In Europe it does exactly what it promises. It keeps a healthy return of 5.2 percent a year, but much steadier, with a volatility of only 9.1 percent and the smallest worst fall of any European strategy, 26 percent. In the United States the same cleaning leaves a return of just 1.6 percent a year and a Sharpe ratio of 0.20, the weakest result on the whole board apart from its own plain version. Europe behaved as the textbook says. The United States did not.

6

The Surprise

Cleaning the signal helped Europe but hurt the United States.

This is the opposite of what was supposed to happen. The whole point of removing the market is to make the signal cleaner and therefore better. In Europe it did. It raised the reward for risk and roughly halved the worst fall. In the United States it did the reverse. It stripped the return down to almost nothing and left a strategy that barely pays you for the risk you are carrying.

The difference is too large to ignore. The same method, using the same forces, improved momentum in one market and damaged it in the other. Something has to explain that. The rest of this study tests three explanations, one at a time, in the order a careful person would try them. The first and most obvious is that the United States simply has more years of data than Europe. We test that next.

7

First Test: Is It Just the Extra Years?

The United States history is longer. Could that alone be the cause?

The most obvious explanation is the difference in the length of the two histories. The United States data run from 1990. The European data effectively begin in 2006, because residual momentum needs three years of history before it can produce its first score. So the United States carries an extra stretch of the 1990s, which was a very good decade for momentum, plus more booms and busts along the way. Maybe that extra, kinder history is doing all the work.

There is a clean way to check. We cut both markets down to the exact months they share, from 2006 to 2018, and measure again. This is what we mean by the matched window: the same set of calendar months for both markets, so the comparison is finally fair. Because Europe already lives almost entirely inside those years, the cutting falls almost entirely on the United States, which loses its earliest and best years. We expected this to help the United States. If its weakness was only caused by having more, kinder years, then taking those years away should pull its result back up towards Europe.

Table 3. **The same strategies, now over the shared years, 2006 to 2018**

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
United States, plain	-3.8%	25.6%	0.00	81.5%
United States, residual	-7.1%	20.4%	-0.23	74.0%
United States, volatility targeted	4.4%	13.1%	0.40	35.3%
Europe, plain	12.8%	24.9%	0.64	74.1%
Europe, residual	5.2%	9.1%	0.60	26.2%
Europe, volatility targeted	13.2%	13.5%	0.99	27.3%

Key observation. *The United States numbers fall sharply, while the European numbers barely move. European residual momentum is exactly the same as before, because its full history already was these years.*

The result is the opposite of what we expected. Cutting the United States back to the same years did not help it. It made it worse. Plain momentum in the United States dropped to a Sharpe ratio of zero, and residual momentum, which was merely weak before, turned firmly negative at minus 0.23. Europe barely changed, because for Europe almost nothing was removed. So having more years of data was not the cause. The clearest way to see this is to put each market's full history next to its matched window, side by side.

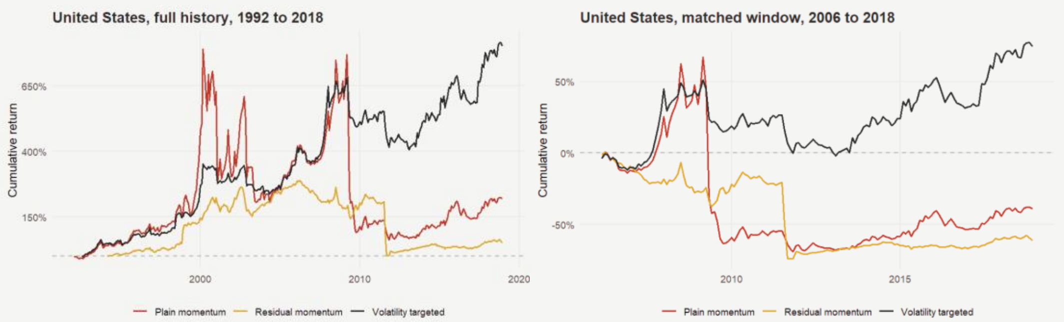

Figure 1. **Cumulative return chart. United States: full history on the left, matched window on the right.** Each line shows what one unit of money grows into over time. The dashed line in the middle is the starting point: above it means a profit, below it means a loss.

These two pictures are the same three United States strategies, and they look almost nothing alike. On the left, over the full history, all three lines end above the starting line, so all three made money in the end, although the residual line is much weaker than the others and fell sharply more than once along the way. On the right, over the matched years only, the picture changes completely. The plain and residual lines drop and finish well below the starting line, so they lost money, and only the volatility targeted line stays up. Nothing about the strategies changed between the two pictures. Only the years did. The healthier picture on the left was being held up by the 1990s and early 2000s, the very years Europe never had. Take those years away, and what is left is the picture on the right, where United States momentum loses money.

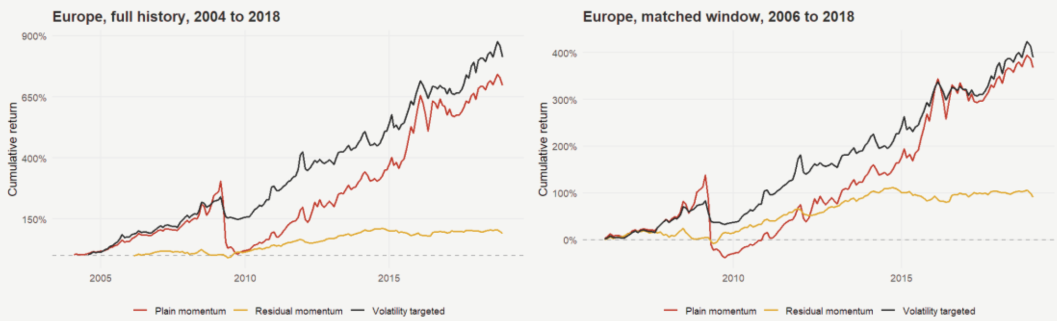

Figure 2. Cumulative return chart. Europe: full history on the left, matched window on the right. The same growth of one unit of money, for Europe, whose data already begin close to the start of the shared years.

For Europe the two pictures tell almost the same story, which is exactly what we would expect. Europe's data already begin around 2006, so cutting to the shared years removes very little. On the left, over the full history, the plain and volatility targeted lines climb strongly and the residual line rises gently but steadily into profit. On the right, over the matched years, the shape is almost identical, just a little shorter, and the residual line is essentially unchanged. Where the United States was transformed by matching the years, Europe was barely touched.

Putting the two markets next to each other makes the point plain. If the difference between them were only about which years were included, then matching the years would have changed both markets. Instead it rewrote the United States and left Europe almost exactly as it was. We can also see the same effect across all six strategies at once in two bar charts, where each strategy gets two bars: one for its full history and one for the matched years.

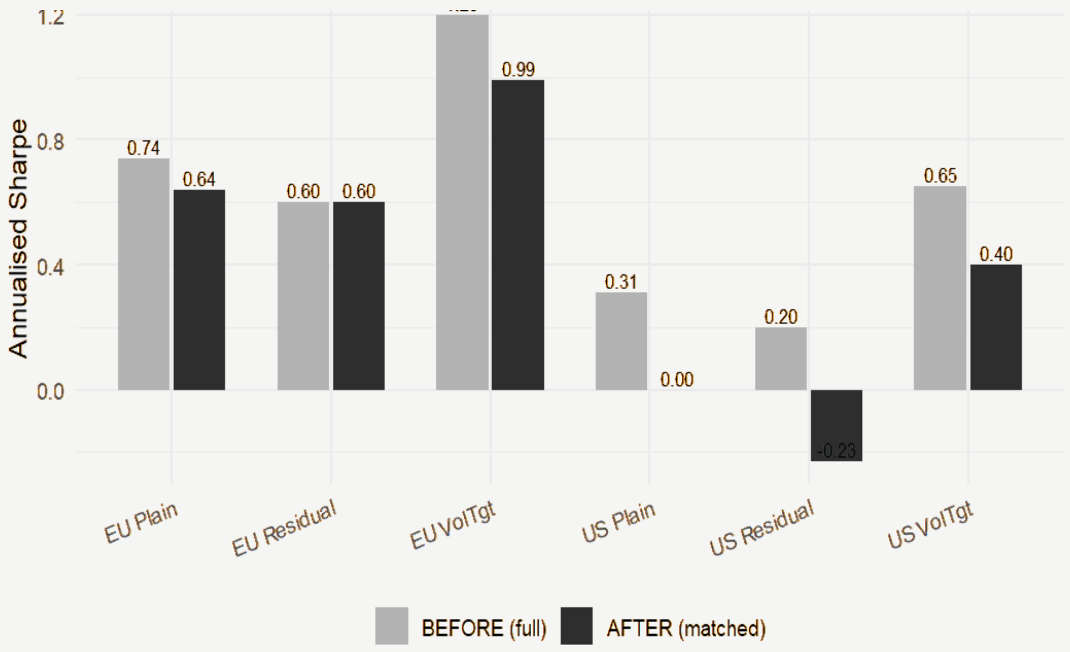

Figure 3. Grouped bar chart of the Sharpe ratio, full history versus matched years. Each strategy has two bars. The lighter bar is the full history; the darker bar is the matched years. You read it by comparing the two bars in each pair.

In this chart, the European pairs are almost equal in height, so matching the years changed Europe very little. The United States pairs drop sharply, and the United States residual bar falls below the zero line, which means it lost money once the kinder early years were removed. The chart says in one picture what the table says in numbers: matching the years harmed only the United States.

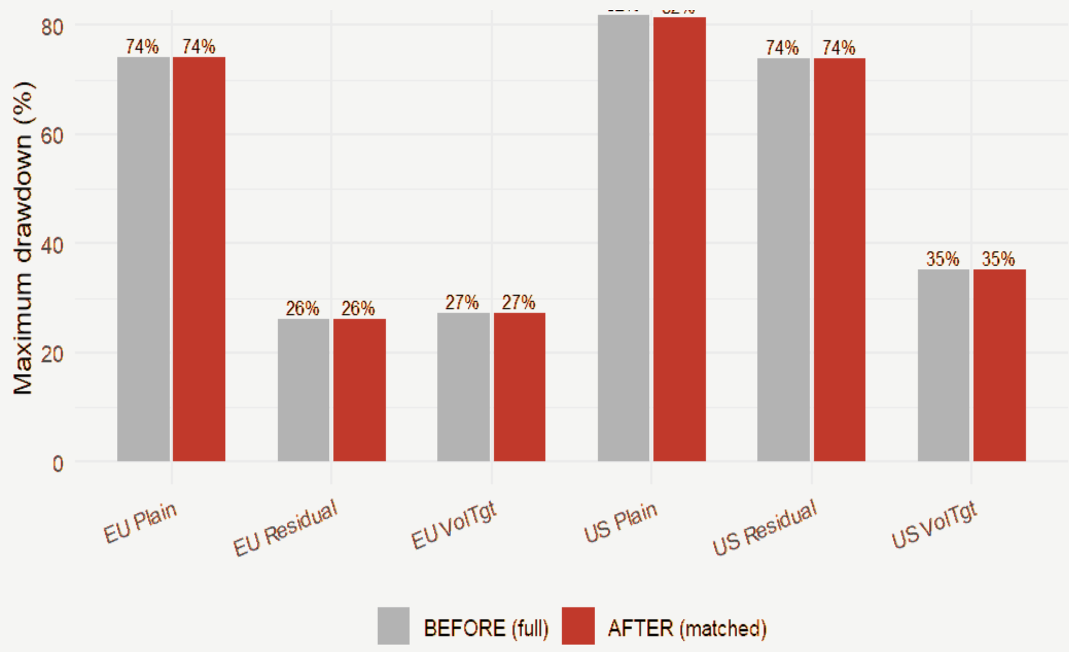

Figure 4. **Grouped bar chart of the maximum drawdown, full history versus matched years.** Again two bars per strategy, the full history and the matched years, with shorter bars meaning a gentler worst fall.

Here almost nothing moves. The worst falls are about the same before and after matching, in both markets. The reason is simple: every strategy's deepest fall happened during the 2008 crisis, and 2008 is inside the shared years. Cutting the early years away removed good years, not the crash. So the crashes were never going to change, and they do not. The conclusion of this first test is clear. The United States weakness is not caused by having more years of data. If the period is not the reason, the next thing to check is the market itself.

8

Second Test: Is It the Market Itself?

Once the market is removed, what is left, and can it be trusted?

Ruling out the years sharpens the question. If the difference is not about the period, perhaps it is about the market. The cleaned up version gives us a direct way to look, because the only difference between plain and residual momentum is whether the market's push is left in or taken out. Comparing the two therefore measures exactly what the market was contributing in each place. We expected that if the United States companies were genuinely strong on their own, then removing the market should leave a smaller but still positive return, just as it does in Europe.

Table 4. **Yearly return with the market left in, and with it removed, over the shared years**

	United States	Europe
Plain momentum, market left in	-3.8%	12.8%
Residual momentum, market removed	-7.1%	5.2%
The difference, what the market was adding	3.4%	7.6%

What changes after removing the market? *The difference between the two rows is roughly how much the market was contributing. In Europe a healthy return remains once the market is removed; in the United States the return falls further into loss.*

Two things stand out. In Europe, removing the market still leaves a healthy 5.2 percent a year, so the European companies carry the signal on their own. In the United States, removing the market lowers the return from minus 3.8 to minus 7.1 percent, so the companies on their own are a drag. The difference row adds a detail that runs against the obvious story. The market actually gave more to Europe, 7.6 percent, than to the United States, 3.4 percent. So the United States problem is not a generous market hiding weak companies. Even with the market's smaller help, the United States signal lost money, and without that help it lost more.

Return alone does not tell the whole story, though. A strategy can make money while still being unreliable, and it can lose money while still being steady. So the next table asks a different question: not how much the United States signal earned, but whether it was steady enough to trust from one month to the next.

Table 5. **How dependable the cleaned up signal was, over the shared years**

	United States	Europe
Share of months that ended positive	51.3%	61.7%
Typical swing from one month to the next	5.9%	2.6%
Worst single month	-48.0%	-6.8%

What these numbers tell us. *A signal that is positive only about half the time is close to a coin toss. A larger month to month swing, and a deeper worst month, both mean a less dependable signal.*

In the United States the cleaned up signal was positive in only 51.3 percent of months. That is barely better than tossing a coin. It also swung more than twice as much from month to month as the European signal, and in its worst single month it fell by 48 percent, almost half its value, against a worst European month of under 7 percent. A signal that has supposedly been cleaned of market noise, yet is still close to a coin toss and can lose half its value in one month, is hard to treat as real information. Both the size and the steadiness of the United States signal point the same way, and away from the calendar.

So far these figures describe what happened. A fair question is whether they could simply be the result of chance. To check, we use a **significance test**. The idea is simple. We begin by assuming the boring position, that nothing special is happening, which is called the **null hypothesis**. We then ask how likely our result would be if that boring position were true, a likelihood called the **p value**. A small p value, below 0.05, means the result would be very unlikely by chance, so we treat it as real. A large p value means the result could easily be luck, so we cannot claim anything. We ran two tests on the shared years.

The first test asks whether the share of positive months is really different from a coin toss. The null hypothesis is that the signal is positive half the time, purely by chance. For the United States the share was 51.3 percent, with a p value of 0.81. That is far above 0.05, so we cannot reject the coin toss: the United States signal is, in statistical terms, **a coin toss**. For Europe the share was 61.7 percent, with a p value of 0.005, well below 0.05, so Europe is positive more often than chance would allow.

The second test asks whether the average monthly return is really different from zero. The null hypothesis is that the true average is zero, meaning no real profit. For the United States the average was minus 0.39 percent a month, which could not be told apart from zero, with a p value of 0.41. For Europe the average was plus 0.45 percent a month, a genuinely positive figure, with a p value of 0.03.

Together these two tests confirm the picture in formal terms. Europe's signal is **statistically real**: it is positive more often than chance, and it earns a return that is genuinely above zero. The United States signal is **statistically**

indistinguishable from noise: it is as likely to be positive as a coin toss, and its return cannot be told apart from zero. The description and the formal test now agree. One market has a real signal, and the other has noise.

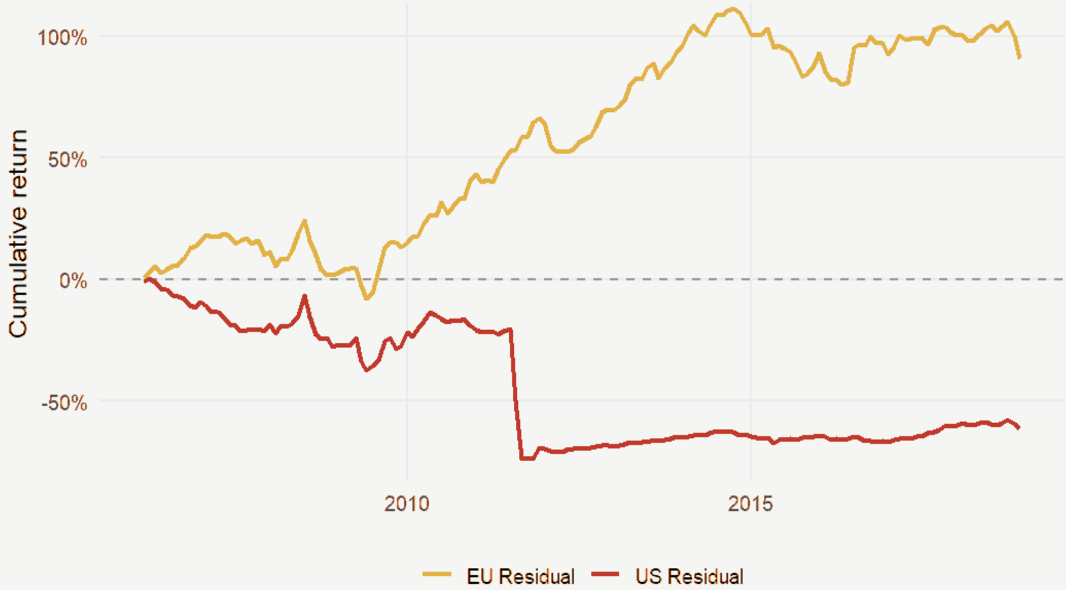

Figure 5. Cumulative return chart. The cleaned up signal in both markets, over the shared years. Growth of one unit of money for the residual strategy alone, in each market, on the same set of months.

This last picture shows the two cleaned up signals side by side over the same years. The European line climbs steadily into profit, while the United States line sinks and stays there. The same method, over the same months, ends in opposite places. That points to a real difference between the two markets, not to anything about the time period. So the second test gives a firm answer: the difference is structural, something in the market itself. One question remains, and a careful reader would ask it. What if we removed the market in too simple a way?

9

Third Test: Is It the Way We Removed the Market?

We removed three forces. What if we remove five?

So far, removing the market has meant removing three broad forces: the market, company size, and value. This three force recipe is the standard one, known as the Fama and French three factor model. But it is not the only choice. A fuller recipe, the five factor model, removes those same three and adds two more. A fair challenge to our result is this: maybe three forces was too rough a clean, and some leftover junk in the United States is what behaved badly. If we clean more thoroughly with five forces, does the United States signal recover?

The two extra forces in the five factor recipe are these;

Profitability is the tendency for companies with strong, steady profits to behave differently from companies with weak profits.

Investment is the tendency for companies that grow cautiously to behave differently from those that expand aggressively.

Removing these two as well leaves a cleaner leftover than before. We expected that the stronger clean might rescue the United States, and we tested it on the matched years, the ones that matter for our conclusion.

Table 6. Removing three forces versus five forces, over the shared years

Cleaned up signal	Annualised return	Sharpe ratio	Worst fall
United States, three forces removed	-7.1%	-0.23	74.0%
United States, five forces removed	-2.3%	-0.03	54.0%
Europe, three forces removed	5.2%	0.60	26.2%
Europe, five forces removed	3.7%	0.44	30.0%

What changes after a stronger clean? *Removing two extra forces lifts the United States, but not above zero. Europe slips a little, because it was already clean and the extra removal trims some real signal away.*

The stronger clean does help the United States. Its return rises from minus 7.1 to minus 2.3 percent a year, its worst fall shrinks from 74 to 54 percent, and its Sharpe ratio improves from minus 0.23 to minus 0.03. So part of the United States weakness really was the rougher three force clean leaving some junk behind. But notice where it ends. Minus 0.03 is still not a winning signal. After a fuller, fairer clean, the United States residual still earns essentially nothing. Europe moves the other way, from 0.60 down to 0.44, and it is worth being precise about what that drop means. When removing the two extra forces lowers Europe's result, it shows that part of Europe's edge was being paid for exposure to profitability and investment, which a five factor view treats as known, rewarded risk rather than pure stock selection. That does not weaken the comparison. If anything it strengthens it, because even after those forces are removed from both markets, Europe stays clearly positive at 0.44 while the United States stays below zero at minus 0.03.

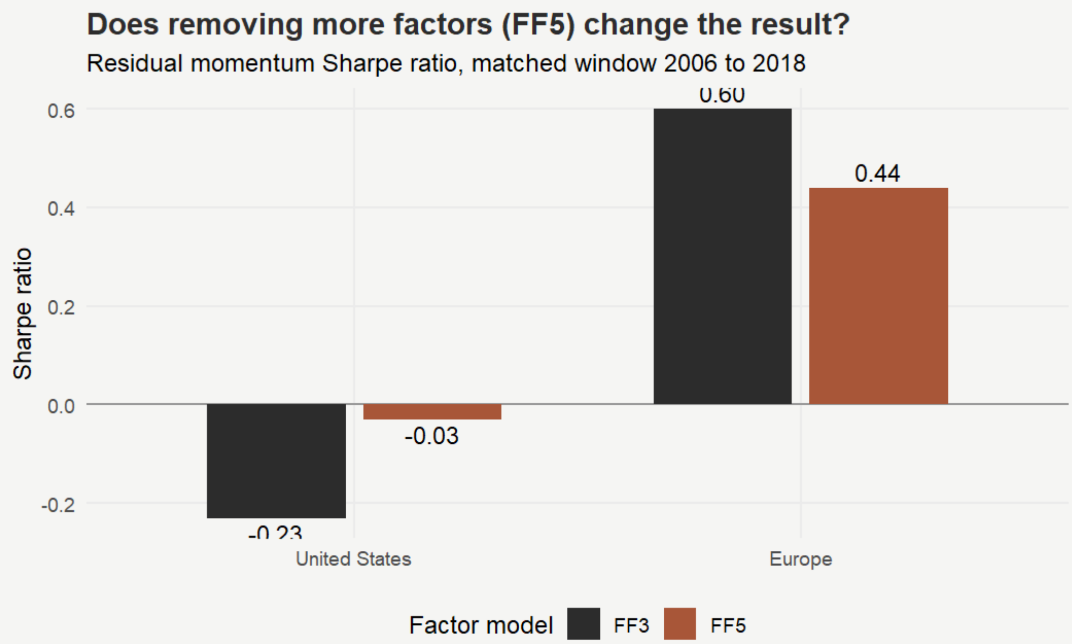

Figure 6. **Bar chart of the Sharpe ratio under each recipe, over the shared years.** Two bars for each market: the darker bar removes three forces, the lighter bar removes five. Above the zero line is a winning signal, below it is a losing one.

The picture makes the answer plain. Both United States bars sit at or below the zero line, while both European bars sit well above it. Removing more forces lifts the United States bar, but not over the line. This is the outcome that makes the finding solid. We removed the market in two different ways, the standard three forces and the fuller five, and the United States signal failed both times, while the European signal stayed positive both times. So the United States weakness is not an accident of one particular recipe. It survives a stronger clean. The difference is not the time period, and it is not the method. It is the market.

10

What It All Means

Three tests, one consistent answer.

The three tests point the same way. Matching the years made the United States worse, not better, so the difference is not the time period. Removing the market left a losing, unreliable signal in the United States and a healthy one in Europe, so the difference is in the market. And removing the market more thoroughly did not rescue the United States, so the difference is not the recipe either. When the same method, over the same months, cleaned the same way, leaves a clean profit in one market and a coin toss in the other, the difference has to live in the markets themselves.

Several reasons could explain why the two markets differ, and our data cannot yet separate them, but they are worth naming because they are related. The leading suspect is concentration. The United States market is now dominated by a small number of very large technology companies, such as Apple, Microsoft, Nvidia, Amazon, and Meta, and when a few giants move together they leave less genuine company by company variety for a residual signal to work with. This is the first thing a careful reader should suspect, and testing it directly is the obvious next step. The market may also simply explain more of what moves United States shares, so that removing it throws away more useful information there. The United States momentum trade may have grown crowded, with many investors chasing the same shares, which tends to make any strategy fragile. And it is possible that the market's push is itself part of what makes momentum work in the United States, so that removing it removes part of the reward rather than cleaning the signal.

What we can say with confidence is narrower and careful. Over the years we studied, removing the market kept momentum healthy in Europe but weakened it in the United States, and that stayed true after we matched the years and after we cleaned more thoroughly. The lesson is that market related information should not be treated as noise everywhere by default. In at least one large market, it appears to carry part of the reward.

11

Limitations

What this study cannot claim.

This study covers one stretch of time, from 2006 to 2018, because that is the span the data allow. The findings describe that stretch. They cannot, on their own, say what will happen in years outside it.

The shared window is also fairly short, at 154 months. A shorter window means less certainty around any single number, so small differences should not be read too strongly, even though the main gap between the two markets is large.

The strategies are also frictionless. They ignore the cost of trading, the difficulty and cost of selling shares short, and the trouble of trading the smallest companies. These frictions would lower the returns of all six strategies, and most of all the plain version, which trades a great deal. They would not obviously favour one market over the other, but they are real and they are not in the numbers here.

Finally, the study uses two broad indices from one data source. A different set of companies, or a different provider, might shift the numbers somewhat. The central contrast between the two markets is large enough that small changes are unlikely to remove it, but it is worth stating plainly.

12

Future Work

The questions this study opens.

The most useful next step is to bring the data up to date. The sample ends in 2018, and the United States market has changed a great deal since then. It would be valuable to test whether United States residual momentum has recovered in more recent years, especially because one would expect the deepest and most studied market in the world not to stay the weaker of the two for long.

With current data in hand, the result should also be put through a fuller set of formal tests than this study runs. The most useful would be these. First, a test of whether the gap between the two markets is statistically significant in a direct comparison, supported by bootstrap and Monte Carlo methods that re-run the result thousands of times to check it is not an accident of one particular sample. Second, cross sectional regressions that control for industry and company size, to test directly whether the dominance of a few very large technology companies is what starves the United States signal. Third, the same tests repeated over shorter sub periods, to see whether the United States weakness is steady through time or driven by one or two unusual stretches. Together these would move the finding from a clear pattern in one sample to a result that has been stress tested from several directions, and would allow a confident claim that residual momentum is not the same in every market.

Next step is to find the exact structural cause. This study shows the difference is structural without proving which structure is responsible. A natural follow up would split the United States result by industry and by company size, to test directly whether the dominance of a few very large technology companies is what starves the signal.

Another step is to widen the comparison. Running the same test in other markets, such as Japan and the larger emerging markets, would show whether the United States is a lone exception or one case of a broader pattern in which the value of removing the market depends on the market itself.

Last step is to manage the problem rather than only describe it. One could remove only part of the market instead of all of it, or combine the cleaned up signal with volatility targeting, to test whether the instability found here can be tamed into something useful.

13

Conclusion

When a cleaner signal is not the better one.

We set out expecting the cleaned up version of momentum to be the better one, as the literature suggests. Europe agreed. The United States did not. The same cleaned up signal that behaves well in Europe was weak in the United States, and the gap was large enough to demand an explanation.

We first suspected the United States simply had more years of data, so we cut both markets to the same years. The gap did not shrink. It grew, and the United States signal turned negative. So the time period was not the reason.

We then removed the market entirely and looked at what was left. In the United States the leftover lost money and behaved close to random, with a worst month of nearly minus 48 percent. In Europe the leftover stayed healthy. So the difference is structural, something in the market itself.

We then asked whether we had removed the market too simply, and removed five forces instead of three. The United States improved but still did not make money, while Europe still won. So the difference is not the method either.

The conclusion is that residual momentum is not always an improvement. Whether removing the market helps or harms depends on the market it is used in. In the United States, over these years, it harmed. Why exactly, whether through the dominance of a few giant companies, a crowded trade, or the market itself carrying part of the signal, is the natural question for the work that follows.

14

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The work this study builds on.

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